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To cite this article: Giuliano Di Baldassarre, Luigia Brandimarte & Keith Beven (2016) The seventh facet of uncertainty: wrong assumptions, unknowns and surprises in the dynamics of human–water systems, Hydrological Sciences Journal, 61:9, 1748-1758, DOI: [10.1080/02626667.2015.1091460](https://doi.org/10.1080/02626667.2015.1091460)

To link to this article: <https://doi.org/10.1080/02626667.2015.1091460>



Published online: 12 May 2016.



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The seventh facet of uncertainty: wrong assumptions, unknowns and surprises in the dynamics of human–water systems

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ABSTRACT

The scientific literature has focused on uncertainty as randomness, while limited credit has been given to what we call here the “seventh facet of uncertainty”, i.e. lack of knowledge. This paper identifies three types of lack of understanding: (i) *known unknowns*, which are things we know we don’t know; (ii) *unknown unknowns*, which are things we don’t know we don’t know; and (iii) *wrong assumptions*, things we think we know, but we actually don’t know. Here we discuss each of these with reference to the study of the dynamics of human–water systems, which is one of the main topics of Panta Rhei, the current scientific decade of the International Association of Hydrological Sciences (IAHS), focusing on changes in hydrology and society. In the paper, we argue that interdisciplinary studies of socio-hydrological dynamics leading to a better understanding of human–water interactions can help in coping with wrong assumptions and known unknowns. Also, being aware of the existence of unknown unknowns, and their potential capability to generate surprises or black swans, suggests the need to complement top-down approaches, based on quantitative predictions of water-related hazards, with bottom-up approaches, based on societal vulnerabilities and possibilities of failure.

ARTICLE HISTORY

Received 16 July 2014
Accepted 2 September 2015

EDITOR

D. Koutsoyiannis

ASSOCIATE EDITOR

S. Weijs

KEYWORDS

Epistemic uncertainty;
socio-hydrology; black
swans; disaster risk
reduction; resilience

Introduction

The number seven has been very popular and widely used for a long time: the Seven Deadly Sins, the Seven Wonders of the World, the Seven Hills of Rome, the Seven Dwarves and the much more recent Seven Little People of Murakami’s *IQ84*. Mandelbrot (1997) introduced the seven states of randomness, and the number seven also has some popularity among hydrological scientists dealing with uncertainty: Pappenberger and Beven (2006) discussed seven reasons not to use uncertainty analysis, while Juston *et al.* (2013) provided seven reasons to be positive about uncertainty.

Uncertainty is usually associated with the throwing of dice, which, strangely enough, have only six (and not seven) facets! A die was used, for instance, as the official logo of the 2013 Leonardo Conference in Kos, Greece, titled “Facets of Uncertainty”. We think that this is consciously (or subconsciously) due to the fact that uncertainty is often directly (or indirectly) related to the concepts of randomness and probability.

Sources of uncertainty can be classified in a variety of ways (Knight 1921, Ferson and Ginzburg 1996, Apel *et al.* 2004, Beven 2012). These classifications enable a better exploration of the different sources of

(unreducible or reducible) uncertainty and contribute to explicitly recognizing the limitations of our analytical frameworks. It has been shown, for instance, that scientists and experts have a tendency to overconfidence, as we tend to grossly underestimate uncertainty (Cooke 1991, Shlyakhter *et al.* 1994) across a variety of studies, experts and questions (Lin and Bier 2008). Differentiating the sources of uncertainty also supports the selection of appropriate methods (e.g. probabilistic or fuzzy) to deal with uncertainty and support the decision-making process (Klir 2006).

In the 1920s, Frank Knight proposed a differentiation between the uncertainties that can be treated as probabilities and what he called the “true uncertainties” that cannot be quantified in probabilistic terms (Knight 1921). More recently, Ferson and Ginzburg (1996) demonstrated the importance of distinguishing between variability and ignorance, as their proper assessment requires different methods. In this paper, we refer to a similar classification, which differentiates between aleatory and epistemic uncertainties (Apel *et al.* 2004, Beven and Smith 2015).

Aleatory comes from the Latin *alea*, which means a die or game of dice. Aleatory uncertainty is related to

the random variability of processes (Koutsoyiannis 2010, Montanari and Koutsoyiannis 2012). Probabilistic methods are valuable tools for dealing with the uncertainty related to chance and randomness, which, at the current time, cannot be reduced by improving our knowledge of the systems through scientific efforts. For instance, flood risk is often estimated by referring to the expected annual flood damage (Arnell 1989). However, the actual damage (direct and indirect, tangible and intangible; see e.g. Giupponi *et al.* 2014, Haraguchi and Lall 2015) of future flood events will significantly depend on unpredictable factors, such as the exact time of the day, and day of the week, when the big, extreme flood event eventually occurs (Di Baldassarre *et al.* 2009a). For instance, the same flood event would result in significantly different damage if it occurs on Sunday night during summer or during rush hours on a Friday afternoon. While the exact time of occurrence of future flood events cannot be deterministically predicted, this intrinsic uncertainty can be assumed to be predominantly aleatory and can easily be treated in probabilistic terms.

Epistemic comes from the Greek ἐπιστήμη, which means knowledge. Sources of epistemic uncertainty are related to the lack of knowledge (Beven and Young 2013). In some instances, we may understand some essential processes and be able to elaborate a number of stylized facts (Kaldor 1957), but we do not have adequate or sufficient knowledge of all the details needed to properly capture these essential processes into our analytical frameworks.

As epistemic uncertainty is not about the game of dice, here we call it the “seventh facet of uncertainty” and discuss its role in the observation and modelling of human–water systems, which is one of the main topics of Panta Rhei, the new IAHS Scientific Decade, dealing with changes in hydrology and society (Montanari *et al.* 2013). In particular, as social dynamics and their interplay with hydrological changes are largely

unknown, and surprises might play a major role, we posit that the study of human–water systems will require going beyond current approaches, whereby epistemic uncertainty is neglected or treated as if it can be represented as aleatory random variability, and with heavy reliance on quantitative predictions. Epistemic uncertainty may not be so simple: it may have complex non-stationary properties that are not easily represented by a simple probability model, and it can be rather arbitrary in its occurrences, resulting in surprises (e.g. Beven 2015).

Our discussion is structured by differentiating three types of lack of understanding, with reference to the study of human–water systems: (i) *known unknowns*, which are things we know we don’t know; (ii) *unknown unknowns*, which are things we don’t know we don’t know; and (iii) *wrong assumptions*, things we think we know, but we actually don’t know.

Human–water systems

Societies strongly rely on access to water resources, which are essential to support livelihoods and provide favourable conditions for socio-economic development. While benefiting from water services, humans also alter the hydrological regime (Koutsoyiannis *et al.* 2009, Di Baldassarre *et al.* 2010a, Vörösmarty *et al.* 2010, Wagener *et al.* 2010, Lane *et al.* 2011). Savenije *et al.* (2014) identified four main types of human impacts on hydrology: (i) direct diversion of water flows (water supplies to cities, industries and agriculture); (ii) stream network transformation (construction of dams and reservoirs); (iii) changing river basin characteristics (deforestation, urbanization, drainage of wetlands and agricultural practices); and (iv) alteration of the regional or global climate (greenhouse gas emissions and land cover changes).

Hence, as societies change the hydrological regime, hydrological changes simultaneously shape societies. Figure 1 shows the coupled dynamics of hydrology

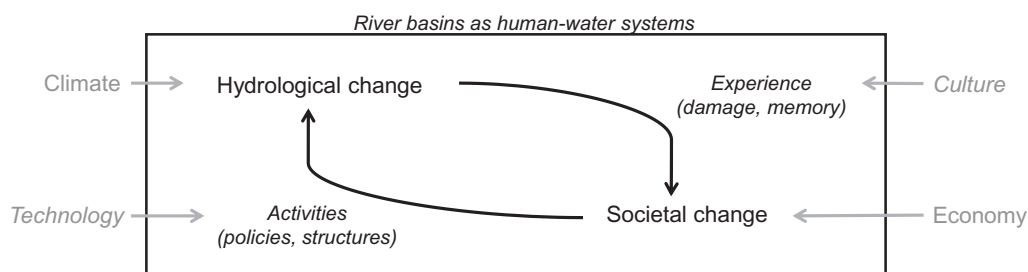


Figure 1. The socio-hydrological cycle. Societies change the hydrological regime via human activities (urbanization, deforestation) and interventions (e.g. dams, levees) while the experience of hydrological changes shapes societies. Human and water systems are deeply intertwined and respond to regional changes in climate, economy, technology and culture.

and society within a river basin driven by changes in climate, economy, technology and culture at a large scale (hereafter called “regional”). For instance, by following the feedback loops of Fig. 1 from the bottom-right, the diagram shows how economic growth can lead to changes in societies, which in turn develop new policies and implement new structures, such as irrigation canals, dams and reservoirs (depending on the available technology) to better exploit water resources and/or cope with hydrological extremes. This process often leads (together with regional climatic changes and variability) to hydrological changes that are then experienced by the same societies, which in turn respond and adapt to these changing conditions.

Socio-hydrological frameworks, such as the one depicted in Fig. 1, are being applied more and more for the interpretation of human–water interactions and feedbacks in many different fields: irrigation, agriculture, droughts and floods (Ostrom 2009, Sivapalan *et al.* 2012, 2014, Srinivasan *et al.* 2012, Montanari *et al.* 2013, Di Baldassarre *et al.* 2013a, 2013b, 2015, van Emmerik *et al.* 2014, Gober and Wheeler 2015, Liu *et al.* 2015, Troy *et al.* 2015).

Recognizing and assessing uncertainty is crucial to provide useful information to decision makers (Pappenberger and Beven 2006, Faulkner *et al.* 2007, Montanari 2007, Blazkova and Beven 2009, Koutsoyiannis 2010, Brandimarte and Di Baldassarre 2012, Beven 2012, 2014, Krueger *et al.* 2012, Juston *et al.* 2013). To this end, a number of probabilistic methods have been developed (e.g. Beven 2009, Di Baldassarre *et al.* 2010b, Neal *et al.* 2013). As they are often based on an (unavoidably) incomplete collection of potential scenarios, the issue arises of what exactly is meant by probability.

Here we argue that the seventh facet of uncertainty plays a major role in the study of dynamic human–water systems. Fully coupled human–water systems are complex and nonlinear (e.g. Liu *et al.* 2007), and the dynamic interplay between hydrology and society is still poorly understood. Thus, the uncertainty affecting the interconnected dynamics of hydrology and society (Fig. 1) is much more complex than the probabilities that e.g. gamblers estimate when playing dice or enjoying casinos. To clarify our point, Fig. 2 shows three examples of time series produced by ludic, water and human–water systems.

The first diagram (Fig. 2(a)) shows a synthetic time series of flood levels, which were randomly generated under the assumption of independent and identically distributed variables and using the mean and variance values of the annual maximum water levels recorded on the Piave River at Ponte delle Alpi (Italy). This is an

example of a time series generated by a “ludic” system. Ludic comes from the Latin *ludus*, which means “play” or “game”. Taleb (2007) introduced the term “ludic fallacy” to refer to the misuse of the narrow world of games, casinos and dice (whereby probabilities are known) to simulate the uncertainty of real-world processes.

Figure 2(b) shows the actual annual maximum water levels recorded at Ponte delle Alpi. This is an example of a time series generated by a water system. Contrary to the randomly generated time series, Fig. 2(b) shows that higher flood levels were observed in the first period. This negative “trend” could be caused by long duration structure, memory and persistence that are often observed in long time series of hydrological data (Koutsoyiannis and Montanari 2007, Blöschl and Montanari 2010, Koutsoyiannis 2013). Probabilistic methods can still be used to handle this type of uncertainty, but they may require the identification of a model of the underlying stochastic structure (Montanari and Koutsoyiannis 2012).

Lastly, Fig. 2(c) shows the entire time series of annual maximum water levels recorded at Ponte delle Alpi. This is an example of a time series generated by a human–water system. The diagram shows the same data as in Fig. 2(b), but it also includes the surprisingly high (and essentially unpredictable) flood level of 9 October 1963, which destroyed the hydrometric station.

An unrepeatable chain of events and cascade of contingencies generated this incredibly high water level. A few years before, upstream of the hydrometric station of Ponte delle Alpi, in the valley of one of the Piave’s tributaries, an artificial lake was created by the newly built Vajont Dam. The filling of the lake during one of the reservoir tests triggered a huge and fast landslide into the lake, which in turn displaced 50 million cubic metres of water. Two giant waves from the lake overtopped the dam and washed away the town of Longarone located downstream (Ward and Day 2011, Delle Rose 2012, Bianchizza and Frigerio 2013, Di Baldassarre *et al.* 2014). Subsequently, this immense volume of water generated a giant flood event that propagated along the Piave River and led to the surprisingly high water level of 9 October 1963, which destroyed the hydrometric station of Ponte delle Alpi (Viparelli and Merla 1968).

While this incredibly high flood level would not be used in traditional hydrological studies that focus on the processes of the water system (Fig. 2(b)), research in dynamic human–water systems should also consider the possibility of generating surprises, such as the one depicted in Fig. 2(c). As human behaviour is difficult

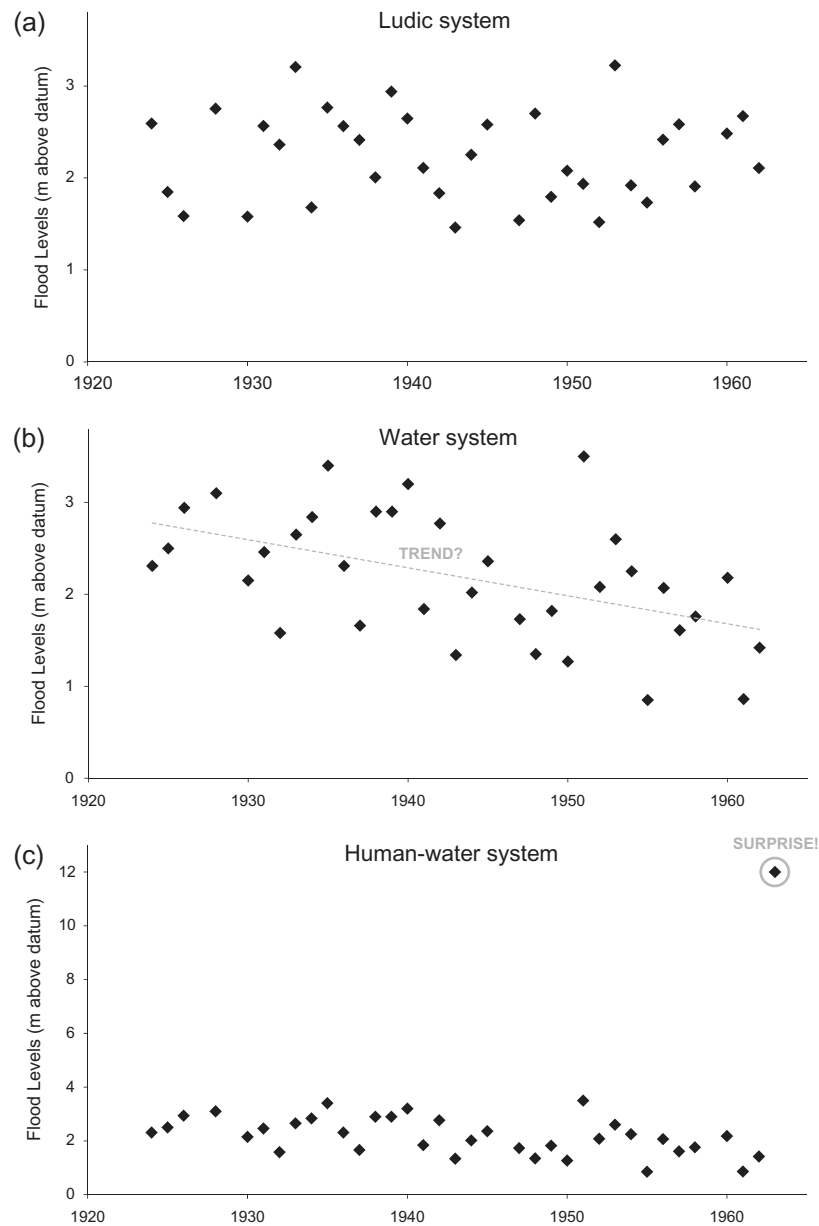


Figure 2. Examples of time series of ludic (a), water (b) and human–water (c) systems. (a) Synthetic time series of flood levels, randomly generated using the mean and variance of the annual maximum water levels recorded on the Piave River at Ponte delle Alpi, Italy. (b) The actual time series of annual maximum water levels. (c) The entire time series of annual maximum water levels, which includes the surprisingly high (and essentially unpredictable) water level of 9 October 1963. From (a) to (c), the seventh facet of uncertainty plays an increasingly major role.

(if not impossible) to predict, here we posit that the seventh facet of uncertainty significantly affects the dynamics of human–water systems. Thus, there is a need to go beyond heavy reliance on quantitative predictions and probabilistic methods for uncertainty assessment.

Known unknowns

Major flooding occurred in Brisbane (Australia) in 2011. More than 10 000 properties were affected and 25 people

died (Bohensky and Leitch 2014). The local population perceived this flood disaster as a surprise, despite the occurrence of major flooding in 1974. People were surprised because, after the 1974 event, a number of flood protection measures were implemented, including the Wivenhoe Dam. The presence of these flood protection structures “led to the popular belief that Brisbane was flood proofed” (Bohensky and Leitch 2014). Similarly, a few years earlier, in 2005, people in New Orleans (USA) were surprised by the catastrophic flooding caused by levee failure during the Katrina event (Kates *et al.* 2006).

New Orleans and Brisbane are only two recent examples of the so-called “levee paradox” (White 1945). The consequence of building (or strengthening) flood protection measures is that the non-experience of frequent flooding reduces societal memory and, as a result, more socio-economic development tends to take place in flood-prone areas. Hence, the reduced probability of flooding might generate increasing potential consequences.

This effect can also be explained by looking at the feedback loops in Fig. 1, starting from the implementation of human activities (levee heightening) that produces hydrological change (reduced frequency of flooding), which is experienced by societies, which then develop more in flood-prone areas.

The levee paradox is an example of a known unknown. The paradox was identified by White (1945) and has been discussed by several authors (Kates *et al.* 2006, Montz and Tobin 2008, Di Baldassarre *et al.* 2009b, 2013a, 2013b, Castellarin *et al.* 2011, Viglione *et al.* 2014). Most flood scientists know about it. Yet methods to capture it in the assessment of future flood risk (which can be defined as a combination of flooding probability and potential adverse consequences) are completely lacking. For instance, when flood defence structures are planned and designed, current methods can indeed estimate the corresponding reduction of flooding probability. However, they do not assess how such a reduction might trigger an increase in the adverse consequences of flooding (Fig. 3).

This is not a trivial aspect in assessing a realistic flood risk. There is evidence that flood risk might even increase in the long term as a result of flood protection measures (see e.g. real-world examples provided by Di Baldassarre *et al.* 2015).

Another example of a known unknown is the “adaptation effect”, which relates to the observation that the occurrence of more frequent flooding is often associated with decreasing vulnerability. This can be attributed to spontaneous learning processes and societal responses due to the experience of hydrological change, e.g. the feedback loops of Fig. 1. For instance, there is empirical evidence that flood damage is often lower when a flood event occurs shortly after a similar one. Wind *et al.* (1999) showed that the losses caused by the 1995 flood on the Meuse River were much lower than those caused by a previous event, of similar magnitude, that occurred in 1993. Moreover, Mechler and Bouwer (2014) recently showed similar dynamics with reference to Bangladesh, and demonstrated the limitations of current analytical frameworks in projecting future disaster risk. Similarly to the levee paradox, this adaptation effect, or learning process, is not captured by the current methods of flood risk assessment.

These effects are not mere paradoxes or exceptions. They actually shape the dynamics of many human–water systems around the world and make many state-of-the-art methods, neglecting them, unrealistic. For example, current methods consistently indicate that decreasing frequency and magnitude of floods would lead to a

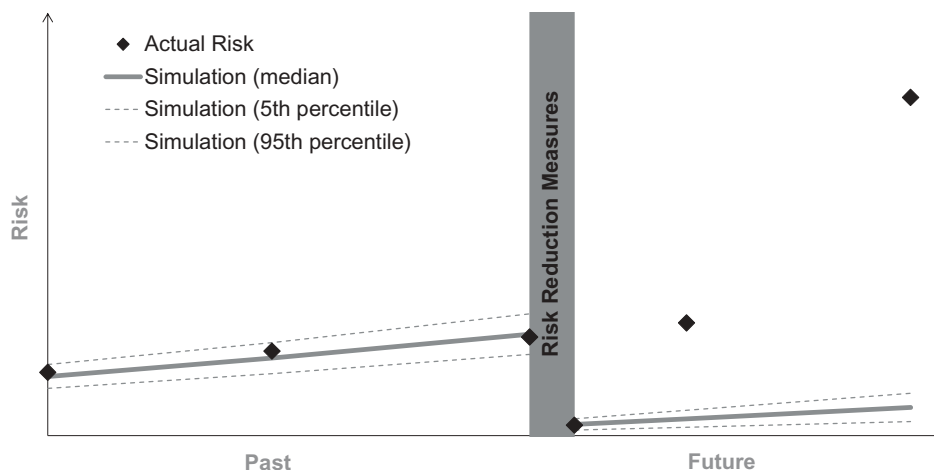


Figure 3. Example of simulating flood risk changes. Current approaches can provide satisfactory results in simulating the reduction of risk immediately after the implementation of structural measures, e.g. decreased probability of flooding due to levee heightening. However, they are unable to capture future changes as they neglect the societal feedback to hydrological change. The diagram shows an example where the actual risk is much higher than the simulated one and well beyond uncertainty bounds, which were computed treating uncertainty as if aleatory. This example reflects the dynamics of risk observed in many real-world cases (e.g. Kates *et al.* 2006, Bohensky and Leitch 2014, Di Baldassarre *et al.* 2015), whereby the reduced probability of flooding has triggered unexpected increases of population and economic activities in flood-prone areas (levee paradox; White 1945).

reduction of risk. However, the levee paradox shows that this would not necessarily be the case. Also, traditional methods consistently indicate that an increasing frequency and magnitude of floods would lead to higher flood risk. However, the adaptation effect proves that this would not necessarily be the case.

Thus, while there has been an enormous development of rigorous, formal, probabilistic methods to estimate uncertainty in flood risk assessment, we still lack fundamental understanding of how flood risk actually evolves in time. It is bizarre that we provide estimates of future flood risk with sophisticated uncertainty bounds, while neglecting crucial dynamics that might determine if flood risk will actually increase or decrease!

Moreover, many puzzles encountered in dealing with water sustainability challenges are caused by our inadequate explanatory power (e.g. water trade paradox, efficiency paradox, peak-water effect; see discussion in Sivapalan *et al.* 2014) of feedbacks between social and hydrological processes (Fig. 1). Sometimes, experts argue that these paradoxes are caused by irrational human behaviour, or mismanagement. But our point is that the study of human–water systems should first aim to understand and simulate how the human–water systems actually work, including spontaneous social dynamics, informalities, values and norms.

The development of new knowledge and the reduction of epistemic uncertainty typically require research efforts that can be classified in five main steps (McDonald 1989):

- (1) Data collection and analysis;
- (2) Examination of these data to determine “salient facts”, which still need a formal explanation;

- (3) Theory development via formulation of models capturing the salient facts;
- (4) Model calibration, validation and uncertainty analysis;
- (5) Application of models to support the decision-making process.

As discussed by McDonald (1989), some scientists are deeply “engaged in work that refines or makes use of the generally accepted model” (steps 4 and 5), while others are “in the process of questioning the generally accepted model” (steps 1–3).

To better understand the behaviour of human–water systems and therefore reduce epistemic uncertainty, we think that research efforts should focus on steps 1–3, as the interplay between hydrological and social processes is still largely unknown. In particular, to develop socio-hydrological theory, there is a need of iterations between historical analyses and formal explanations (e.g. conceptual modelling) of the salient facts, such as the aforementioned levee paradox and adaptation effect.

Historical analyses of human–water interactions over long time scales, i.e. centuries, can be very challenging. For instance, Fig. 4 shows the growth and collapse of the Maya civilization in terms of human population. Many studies have related this to hydrological conditions and shown how technological revolutions triggered population and economic growth in the Maya lowlands, whereas the persistence of drought conditions eventually led to the societal collapse (Haug *et al.* 2003, Gill *et al.* 2007). However, it has been difficult to rigorously test these causal links because the temporal resolution of human population data reconstructed from archaeological records is very coarse (Fig. 4), while the proxies used for dating historical droughts are affected by significant

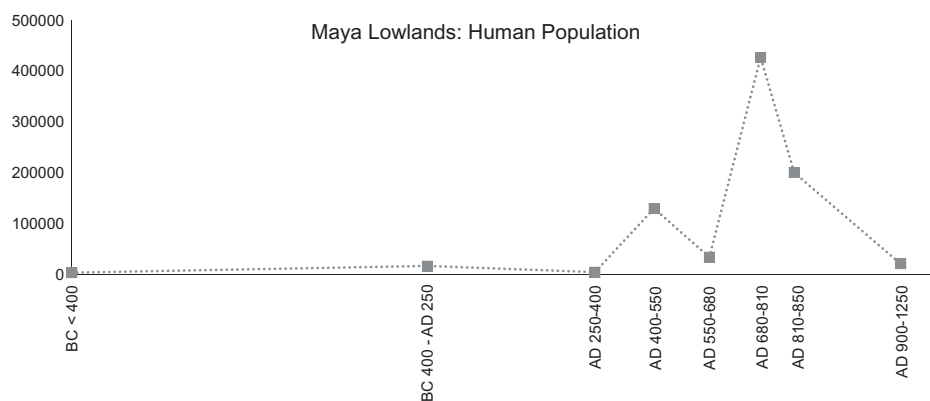


Figure 4. Human population history in the Maya Lowlands. The diagram shows demographic growth periods and collapses, plausibly due to hydrological extremes, i.e. persistent drought conditions (Gill *et al.* 2007).

uncertainties (Aimers and Hodell 2011; see also Yancheva *et al.* 2007, related discussion).

Empirical studies of human–water interactions at time scales relevant for water management and disaster risk reduction, i.e. years to decades, can be much more feasible. Urbanized deltas and floodplains, for instance, are examples of ideal laboratories for the (inter- or trans-disciplinary) study of the interplay between social and hydrological processes as the interactions between human and water systems are apparent and have relevant impacts (Di Baldassarre *et al.* 2013a). In these flood-prone areas, human settlements, flood control measures, and memory of flooding have gradually co-evolved at similar temporal (years to decades) and spatial (floodplain) scales, while they have been also significantly affected by the sudden and localized occurrence of flooding events.

Long time series of demographic, economic and hydrological data along with information about human adjustments to floods are already available for many case studies, such as New Orleans, the Tiber in Rome, and the Dutch delta (Werner and McNamara 2007, Aldrete 2007, De Moel *et al.* 2011). Also, social science literature on water-related disasters (e.g. Ewen 2014) can give useful information about human–water interactions.

Yet, to better understand the dynamics emerging from the (poorly explored) interactions and feedbacks between human and water systems, there is a need to start collecting more empirical evidence. For instance, while the levee paradox had already been identified by White (1945) and been widely discussed in the literature, systematic datasets about its magnitude in real-world case studies are still missing. Conceptual models, i.e. hypotheses of how socio-hydrological systems work (e.g. Di Baldassarre *et al.* 2013b, Viglione *et al.* 2014), can provide insights about the type of data that we need to collect to observe these feedback mechanisms between physical and social processes.

Hence, empirical studies and hypothesis formulation and testing via conceptual models (steps 1–3) can significantly contribute to a better understanding of the dynamics of human–water systems, and therefore reduce this type of epistemic uncertainty.

Unknown unknowns

The study of human–water systems requires an explicit treatment of the interplay between social physical processes. In this context, besides the aforementioned paradoxes that urge more understanding, there are many other things we don't even know we don't know (as stated by Donald Rumsfeld in February

2002). Some of these unknown unknowns may occasionally result in the so-called “black swans”: unexpected events with an extremely high impact on the system, which are essentially impossible to forecast. They are one form of epistemic uncertainty, which is unlikely to be captured in any error distribution derived from an analysis of past data. Yet, after their occurrence we will usually attempt to rationalize and explain them (Taleb 2013). These unexpected events are typically created by unique, unrepeatable combinations of contexts and cascades of contingencies, such as the surprisingly high water level at Ponte delle Alpi (Fig. 2(c)). However, after they occur, humans tend to overreact and overestimate their probability, potentially leading to some unintended consequences.

While, as discussed above, more interdisciplinary research exploring the dynamics of human–water systems can help reduce the epistemic uncertainty related to known unknowns, there is nothing we can do about unknown unknowns as we don't even know what we don't know. However, being aware of their potential occurrence is crucial in supporting the process of water management and disaster risk reduction. In this sense, black swans remind us about the importance of reducing the negative impacts of unexpected events, rather than focusing only on the precise (but most likely inaccurate) estimation of their close-to-zero probability (Makridakis and Taleb 2009a, 2009b). Decreasing the potential adverse consequences of water-related disasters by enhancing the resilience (and reducing the vulnerability) of human societies can be more robust than relying heavily on predictions of the close-to-zero (but essentially unknown) probability of water-related disasters caused by unrepeatable combinations of contexts and cascades of contingencies. For instance, improving evacuation and contingency plans does not necessarily require an accurate and precise estimation of probabilities, but it can significantly increase the resilience of human–water systems, i.e. the ability to recover after an event.

Thus, potential surprises and the impact of epistemic uncertainties suggest the need to go beyond heavy reliance on quantitative predictions and traditional top-down approaches, based on probabilistic assessments of water-related hazards and associated uncertainty. In this context, Blöschl *et al.* (2013) comprehensively discussed the potentialities of complementing traditional top-down approaches with bottom-up ones. Bottom-up approaches do not start from probabilistic predictions, but rather from the societal and economic vulnerability of communities and individuals, and they explore the possibilities (rather than the probabilities) of failure by explicitly

considering the expertise of local stakeholders and risk managers (see e.g. Lane *et al.* 2011). For instance, Wilby and Dessai (2010) showed how the vulnerability of human–water systems to droughts and water scarcity can be reduced by enhancing the connectivity of water supply infrastructures and making abstraction licences time-limited. While this combination of measures could not be considered “optimal”, it is more robust than alternative low-regret options to the potential occurrence of unexpected events with potentially devastating consequences. Thus, it is by acting from a vulnerability’s viewpoint that water managers and scientists can reduce the negative impacts of unknown unknowns and potential surprises.

Wrong assumptions

Besides unknowns, there are also things we think we know, but we actually don’t know. This is what we call here “wrong assumptions” (also sometimes called “ontological uncertainties”). Any scientific work, including hydrological or socio-hydrological studies, is necessarily based on a number of assumptions. Aware of the limitations of our hypotheses, we typically follow a parsimonious approach and focus on the dominant processes that drive the dynamics. However, while some assumptions may work for a number of case studies, they can be significantly wrong in other cases, including assumptions about the nature of the uncertainties involved.

To make this point, we introduce two fictitious characters who are inspired by the ludic fallacy (Taleb 2007): Dr Maria Smith, graduated *summa cum laude* at MIT, the youngest professor at Princeton University. She is a “logical positivist”, who believes in the power of deductive reasoning based on fundamental assumptions. “Smart” Angie, perhaps graduated somehow/somewhere, is a great entrepreneur who became very rich in a few years. She is, on the other hand, a “sceptical empiricist”, who puts more faith in putting the available evidence to robust testing.

A test is made. A coin is flipped 99 times, and each time it comes up heads. The two observers are asked what the odds are that the 100th flip would also come up heads.

Without any hesitation, Dr Smith says: “Odds are not affected by previous outcomes, so the odds must be 50%!”

Smart Angie thinks about it and eventually says: “Well, if it came up heads 99 times in a row there must be something wrong with this coin! So, odds must be much more than 50%!”

The point made by Smart Angie is that the coin must be loaded. In classical terms, odds of the coin

coming up heads 99 times in a row are so low that the *assumption* that the coin had a 50% chance of coming up heads is *most likely wrong*.

Similarly, the occurrence of many 1-in-100 year flood events within a few years may suggest issues with the typical assumption of treating annual maximum flows as time series of independent and identically distributed random variables. Hydrological extremes can be affected by long-term persistence and memory related to climate variability. Blöschl and Montanari (2010) as well as Hall *et al.* (2013) showed examples of occurrence of flood-rich and flood-poor periods (see also e.g. Fig. 2(b)).

This is an example of the many hypotheses that have become standard and have not been sufficiently challenged (similarly to the way Dr Smith keeps assuming the coin to be fair). While the assumption of treating hydrological extremes as independent and identically distributed can be deemed acceptable (from a practical viewpoint) in a few instances, this assumption should be tested and not merely be taken for granted. For another hydrological example see the discussion of preferential flows in relation to the transport of phosphorus and pesticides in Beven and Germann (2013).

When it comes to the nonlinear dynamics of human–water systems, the issue of making fundamentally wrong assumptions becomes even larger because we lack fundamental understanding about how different social groups respond to hydrological changes, as attitudes to risk can strongly vary across societies depending on political and socio-economic conditions as well as cultural values (Kahneman and Tversky 1979, Thompson *et al.* 1990, Viglione *et al.* 2014). And given wrong assumptions, of course, we should expect surprises in the future (as seen also in other disciplines, such as economics).

It should be noted that some wrong assumptions could be related to known unknowns. For instance, while historical changes show that the dynamics of future flood risk can be significantly affected by socio-hydrological feedback loops, these feedbacks are, as discussed above, often neglected in state-of-art assessments of future flood risk (Fig. 3). Moreover, unknown unknowns can also cause some other wrong assumptions as we (might) become aware of their fallacy only after the occurrence of surprises or black swans. So, the three categories used in this paper can potentially overlap.

Conclusions

Despite centuries of water management, we still lack fundamental knowledge of the essential dynamics driving the long-term behaviour of human–water systems. Over the past decades, the focus has been on assessing

in probabilistic terms the uncertainty of the assumed behaviour (how it should work) of the system, rather than exploring ranges of the actual behaviour of the system (how it actually works). This has been related to the focus on uncertainty as randomness along with limited credit to epistemic uncertainty, which here we called the seventh facet of uncertainty.

The increasing impact of human activities on hydrological dynamics, in a time that some call the Anthropocene, has led to a growing interest in the study of water–society interactions (Fig. 1). As the dynamics of human–water systems are still largely unknown, and social dynamics are highly unpredictable, we argued that there is a need to go beyond current approaches (focusing on uncertainty as randomness and probabilistic methods) and give more credit to the seventh facet of uncertainty.

To increase our knowledge of the behaviour of fully coupled human–water systems and reduce epistemic uncertainty, we proposed an iteration of new observations, empirical studies and conceptual modelling (for formulating and testing hypotheses) of coupled dynamics of hydrology and societies. Hence, scientific understanding is believed to be a way to deal with wrong assumptions and known unknowns in the study of the interplay between social and hydrological processes, which is one of the main focuses of Panta Rhei, the IAHS's current scientific decade.

We also discussed that, while unknown unknowns cannot be understood as we don't even know what we don't know, being aware of their existence, and their potential capability to generate surprises or black swans, suggests the need to complement top-down approaches, based on quantitative predictions of water-related hazards, with bottom-up approaches, based on societal vulnerabilities and possibilities of failures.

Acknowledgements

We would like to acknowledge the guest editor Dr Steven Wejis and two anonymous reviewers for providing constructive comments on our paper. The present work was developed within the framework of the Panta Rhei Research Initiative of the International Association of Hydrological Sciences (IAHS).

Disclosure statement

No potential conflict of interest was reported by the authors.

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